SOP: CNC Spindle Thermal Dissipation & Cooling Circuit

5-Axis Spindle Mill Maintenance & Heat Dissipation Failure (HDF) Protocol

ASSET CODE: CNC-M102 **DOC REF:** SOP-CNC-M102-COOL-V4 **STANDARD:** ISO 10791-6 **BAY:** Heavy Machining Bay

1. Subsystem Description & Sensor Operational Limits

The **CNC-M102** milling head incorporates an integrated forced-air and liquid cooling assembly to regulate spindle stator sleeve temperatures under heavy duty cycles:

- **Air Temperature (T_{air}):** Ambient intake sensor at manifold (295 - 303 K).
- **Process Temperature (T_{proc}):** Thermistor embedded in stator coil (305 - 312 K).
- **Nominal Temperature Delta (ΔT):** T_{proc} - T_{air} ≥ 8.6 K under machining load.
- **Critical Threshold:** If ΔT < 8.2 K while Rotational Speed < 1380 rpm, the system triggers **Heat Dissipation Failure (HDF)**.

> **DANGER — HIGH VOLTAGE & STORED ENERGY:** Execute Lockout-Tagout (LOTO) on master breaker `SW-01` before removing fan CF-02 shroud. Wait 15 minutes for DC bus discharge.

2. Disassembly & Inspection Checklist

1. **Isolate Electrical Supply:** Switch master disconnect to OFF and lock with personal technician padlock.
2. **Fan Shroud Removal:** Loosen 8x M6 socket bolts securing fan CF-02 and filter housing.
3. **Filter Debris Cleaning:** Remove stainless mesh filter. Clean accumulated swarf and aerosol sludge with ultrasonic bath or degreasing solvent. If resistance > 120 Pa, replace with Part # `CF-FLT-098`.
4. **Coolant Flow Inspection:** Inspect braided PTFE hoses for constriction or sediment build-up. Confirm reservoir oil is clear (Mobil Velocite No. 6).
5. **Thermal Compound Renewal:** Unclamp RTD sensor, clean with isopropyl alcohol, and apply high-conductivity thermal paste (> 3.5 W/mK).
6. **Reassembly & Baseline Run:** Torque bolts to 9.5 Nm. Run spindle at 1000 rpm for 10 minutes; verify temperature delta stabilizes ≥ 9.0 K.